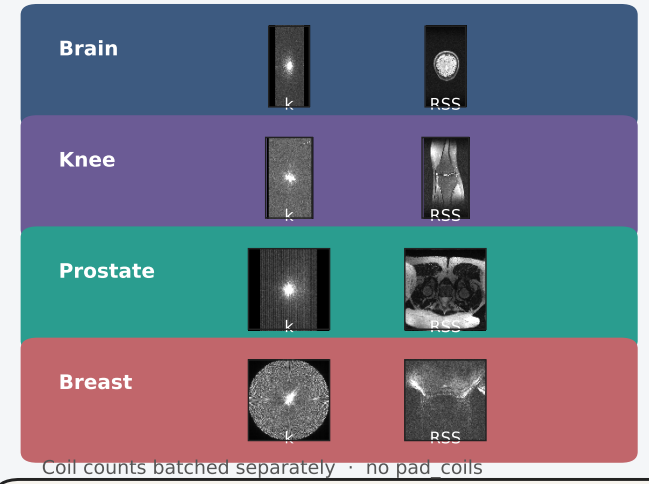
Conditional phase flow matching for multi-anatomy k-space synthesis

Learn $p(\varphi \mid m, \text{organ})$, then $k_c = \text{FFT}(S_c\, m\, e^{i\varphi})$ · $\sum_c |S_c|^2 = 1$

A Fully sampled FastMRI (S_train)

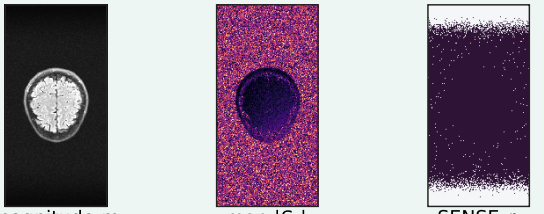
Native FOV · organ-conditioned volumes



Coil counts batched separately · no pad_coils

B Train labels (not ACS / ESPIRiT)

$S_c = \text{IFFT}(k_c)/\text{RSS}$ (FULL_RSS on S_train k)



Unit-phase target

$$x_1 = (\cos\varphi,\ \sin\varphi)$$

SENSE combination after FULL_RSS maps

$$\sum_c |S_c|^2 = 1 \ \Rightarrow\ \text{RSS}(\text{IFFT}(\tilde{k})) = m$$

C OT conditional flow matching

128-ch / 4-pool U-Net · AdaIN(t, organ)

$$v_\theta(x_t, t \mid m, \text{organ})$$

input: x_t + m + t channel

$$x_t = (1-t)\,x_0 + t\,x_1$$

organ: brain, knee, prostate, breast

Training (velocity only)

$$x_0 \sim \mathcal{N}(0, I)$$

$$v^\star = x_1 - x_0$$

$$\mathcal{L} = \| v_\theta - v^\star \|^2$$

Eval: Euler ODE, 30 steps, project to S^1

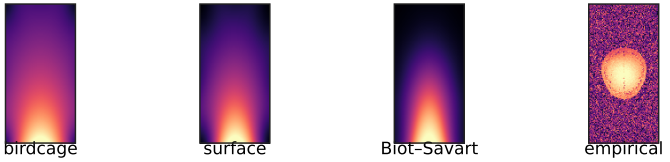
D Inference: phase sample + sensitivity maps + recon transfer

Phase (R_train magnitude; k hidden)

m + organ → x_0 noise → ODE 30 steps → $\hat{\varphi}$ on S^1

Sensitivity maps at apply time (no R_train k)

DIRECT coil simulator: birdcage · surface · Biot–Savart · empirical



Forward model

$m, \hat{\varphi}$ → S_c^{sim} → $\tilde{k}_c = \text{FFT}\left(S_c^{\text{sim}}\, m\, e^{i\hat{\varphi}}\right)$ → synth $\tilde{k}$

Matched recon pair (same R_train patients, same masks; ACS maps at recon time)

Real k · R_train | Synth $\tilde{k}$ · R_train → test both on real k of R_val / frozen R_test

S_train 55% · synthesizer train | S_val 5% · synthesizer val | R_train 40% · recon train | R_val / R_test · official val, real k

Breast (no official val): 45 / 5 / 40 / 5 / 5. Apply-time maps: simulated (birdcage / surface / Biot–Savart) or empirical. Brain stratified by contrast.